

MA Mentored Research: α –Potential Games for Mixed-Autonomy Traffic

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1 Introduction

The deployment of connected and automated vehicles (CAVs) raises a basic control question: how should a large population of autonomous vehicles be coordinated when they share the road with human-driven vehicles (HDVs) whose behavior is unpredictable and non-cooperative? Most existing CAV control schemes fall short on at least one significant criteria. For instance, centralized optimal control schemes assume a shared objective and impose communication and computation costs that do not scale. At the same time, Mean-field game (MFG) approaches, which model the limit of many homogeneous agents interacting through a population aggregate, scale quite well but lose the local pairwise structure that models real driving decisions and assume a homogeneity incompatible with mixed traffic.

Recently, Di et al. (2025) proposed an α -potential game formulation for the pure-CAV problem. The framework reduces equilibrium computation to a single optimization problem, supports finite N , and addresses heterogeneity and strong local interactions. However, the original formulation considers a homogeneous CAV population in isolation; it does not address the mixed-autonomy regime in which CAVs share the road with HDVs.

This research projects asks: Can the α -potential game framework be extended from a pure-CAV setting to a mixed-autonomy setting in which CAVs interact with a macroscopic HDV field, while still admitting a tractable α bound that supports approximate Nash equilibrium computation? Specifically, given a finite population of CAVs whose decisions perturb an HDV density governed by a Lighthill–Whitham–Richards (LWR) type PDE, we ask:

- While maintaining a similar structure to Professor Di’s paper, how should HDV exposure enter the CAV running cost?
- How does a unilateral CAV deviation propagate through the coupled HDV field, and under what regularity conditions can this propagation be controlled?
- What is the resulting α bound on the mixed-traffic game?

The remainder of this report describes our progress on these sections: an overview of the α -potential framework we are building on (Section 2), an initial attempt at incorporating HDV exposure that revealed a structural double-counting issue in (Section 3), the corrected macroscopic-field formulation and resulting α bound (Section 4, 5), and the numerical experiments intended to validate our formulation (Section 6, 7).

2 α -Potential Game Framework

Our work is directly built on the framework introduced in Di et al. (2025).

In the pure-CAV setting, each player $i \in \{1, \dots, N\}$ controls its state $X_{i,t}$ on a horizon $[0, T]$ via the SDE

$$dX_{i,t} = b_i(X_{i,t}, \phi_i(t, X_{i,t})) dt + \sigma_i(X_{i,t}, \phi_i(t, X_{i,t})) dW_t, \quad X_{i,0} = \xi_i,$$

where ϕ_i is player i 's feedback policy. The player's expected cost under joint profile $\phi = (\phi_1, \dots, \phi_N)$ is

$$J_i(\phi) = \mathbb{E} \left[\int_0^T \left(f_i(X_{i,t}^{\phi_i}, \phi_i(t, X_{i,t}^{\phi_i})) + \sum_{j \neq i} \lambda_{ij} K(X_{i,t}^{\phi_i} - X_{j,t}^{\phi_j}) \right) dt + g_i(X_{i,T}^{\phi_i}) \right],$$

combining an individual running cost f_i , terminal cost g_i , and pairwise interactions $\lambda_{ij}K(\cdot)$ that capture local effects such as collision avoidance.

The key idea of an α -potential game is the existence of a single scalar functional $\Phi : \pi \rightarrow \mathbb{R}$ such that, for every player i and every unilateral deviation $\phi_i \rightarrow \phi'_i$,

$$\left| (J_i(\phi'_i, \phi_{-i}) - J_i(\phi_i, \phi_{-i})) - (\Phi(\phi'_i, \phi_{-i}) - \Phi(\phi_i, \phi_{-i})) \right| \leq \alpha.$$

That is, Φ tracks unilateral changes in each player's cost up to a uniform error α . In other words, one can think of the potential function ϕ as a global measure of the change in the game. When one player deviates, holding the action of all other players' fixed, a 'good' potential function ϕ is able to capture the utility/gain of the deviating player through its scalar-value output. This notion of 'good' is represented by the α -parameter; an $\alpha = 0$ represents a perfect potential function, while large α indicates a major discrepancy between the global scale and individual agents. By Proposition 3.1 of Di et al. (2025), any approximate minimizer of Φ is automatically an $(\alpha + \varepsilon)$ -Nash equilibrium of the original game. This collapses the difficult problem of finding an approximate Nash equilibrium of an N -player stochastic game into a single optimization problem in Φ , with the parameter α measuring how far the game is from being an exact potential game.

For the pure-CAV problem, Di et al. (2025) construct Φ explicitly by symmetrizing the pairwise interaction term, and obtain

$$|\Delta_i \Phi - \Delta_i J_i| \leq T \|K\|_\infty \max_{i \in [N]} \sum_{j \neq i} |\lambda_{ij} - \lambda_{ji}|,$$

showing that α is controlled entirely by the asymmetry of the pairwise weights. Their paper introduces a decentralized policy-gradient algorithm that minimizes Φ using per-agent neural-network policies and local gradients. Our project asks how much of this structure survives once HDVs are added to the picture.

3 First Attempt and Double-Counting

My first attempt at extending Di's framework to incorporate density information was to introduce a density-dependent penalty. Before tackling the HDV field directly, I wanted to start by computing local CAV density and finding a way to include it in the cost function, just to build familiarity in the modeling. To do this, we initially defined a Gaussian smoothing kernel K_h centered at vehicle i , computed a local CAV density

$$\rho^{\text{CAV}}(x) = \frac{1}{N} \sum_{j=1}^N K_h(x - X_j),$$

and added a penalty $\Psi_i(\rho^{\text{CAV}}(X_i))$ to each running cost. The intuition was simple: if the area around a vehicle is densely populated, that vehicle pays a higher cost, so it is encouraged to go to less dense areas.

Following the same formulation as in the pure-CAV case, we constructed a candidate potential by aggregating the density penalty across vehicles, and bounded the gap $|\Delta_i \bar{\Phi}^{\text{dens}} - \Delta_i J_i|$ under a unilateral deviation. The complication is that when vehicle i deviates, it changes ρ^{CAV} at every other vehicle's location, so the aggregate density penalty captures more variation than the unilateral change in vehicle i 's own cost. The excess is what we hereinafter call the *density spillover term*, the term we would like to control

$$\eta = \sum_{k \neq i} \left[\Psi_k(\rho^{i'}(X_k)) - \Psi_k(\rho^i(X_k)) \right].$$

Under a Lipschitz assumption on Ψ , we obtained a bound on η and then on α :

$$|\Delta_i \bar{\Phi}^{\text{dens}} - \Delta_i J_i| \leq T \|K\|_{\infty} \max_{i \in [N]} \sum_{j \neq i} |\lambda_{ij} - \lambda_{ji}| + \frac{2TL \|K_h\|_{\infty} (N-1)}{N}.$$

This upper-bound initially seemed promising. However, after consulting with Professor Di, the formulation is **conceptually wrong**. The pairwise interaction term $\sum_{j \neq i} \lambda_{ij} K(X_i - X_j)$ already encodes the influence of nearby CAVs on vehicle i . Adding a CAV-density penalty $\Psi_i(\rho^{\text{CAV}}(X_i))$ on top of the pairwise term double counts the CAV-CAV interaction, as it appears in both the pairwise λK and in Ψ . The bound is still mathematically invalid, but insignificant considering the bound originally derived in Professor Di's paper.

This realization required a complete restructuring of the cost function. We learned that the ρ^{CAV} should not appear as a direct penalty in the CAV cost. Instead, ρ^{CAV} should enter the model only as an input to the HDV exposure, which is what the penalty should be targeted on. This mistake was an important learning experience and resulted in the model's current form.

4 Correctly Incorporating HDV Exposure

The corrected formulation treats HDVs as a continuum field at the macroscopic level while keeping CAVs as a finite-player game at the microscopic level. The CAV running cost still has three components: an individual cost f_i , the original pairwise CAV-CAV interaction $\sum_{j \neq i} \lambda_{ij} K(X_i - X_j)$ from Di's framework, and an HDV exposure penalty $\Psi_i(\zeta_t(X_i))$, where

$$\zeta_t(x) = (\omega * \rho_t^{\text{HDV}})(x)$$

is a kernel-smoothed view of the HDV density at the CAV's location. The smoothing kernel ω determines the bandwidth of how much the HDV field is sampled from the CAVs position.

The HDV density evolves according to an LWR-type conservation law whose flux depends on the total density:

$$\partial_t \rho^{\text{HDV}, \phi}(x, t) + \partial_x \left(\rho^{\text{HDV}, \phi}(x, t) V_H(\rho^{\text{HDV}, \phi}(x, t) + \rho_t^{\text{CAV}, \phi}(x)) \right) = 0,$$

where V_H is a fundamental-diagram velocity (we use the Greenshields form $V_H(\rho) = v_{\max}(1 - \rho/\rho_{\text{jam}})$ in our numerical experiments) and $\rho_t^{\text{CAV}, \phi}$ is the CAV density obtained by kernel-smoothing the

CAV trajectories under joint profile ϕ . The resulting system couples a finite-player CAV game to a PDE-driven HDV field.

We considered two regimes for the HDV field. In the independent regime, the HDV density evolves on its own and is exogenous to the CAV profile ϕ . In this case a unilateral CAV deviation does not perturb the HDV field, the aggregated HDV penalty term cancels in the deviation calculation, and we recover the Di bound exactly. This case is mathematically correct, but largely uninteresting because a realistic mixed-traffic model must have the HDV field respond to CAV behavior. This is addressed in the dependent regime, where $\rho^{\text{HDV},\phi}$ is the unique solution of the LWR equation induced by ϕ through the flux. The remainder of this report focuses on the dependent regime.

5 α Bound for the Coupled CAV–HDV Game

5.1 Candidate potential and decomposition

Building on Di et al. (2025), we propose a candidate potential functional

$$\Phi^{\text{dep}}(\phi) = \mathbb{E} \left[\int_0^T L^{\text{dep}}(t) dt + G(X_T^\phi) \right],$$

where

$$L^{\text{dep}}(t) = \sum_{k=1}^N f_k(x_k, a_k) + \sum_{k=1}^N \Psi_k(\zeta_t^\phi(x_k)) + \sum_{1 \leq k < \ell \leq N} \frac{\lambda_{k\ell} + \lambda_{\ell k}}{2} K(x_k - x_\ell), \quad G(x) = \sum_k g_k(x_k).$$

Computing $|\Delta_i \Phi^{\text{dep}} - \Delta_i J_i|$ under a unilateral deviation gives error sources:

$$|\Delta_i \Phi^{\text{dep}} - \Delta_i J_i| \leq \underbrace{T \|K\|_\infty \max_{i \in [N]} \sum_{j \neq i} |\lambda_{ij} - \lambda_{ji}|}_{\text{pairwise asymmetry}} + \underbrace{\eta_i^{\text{HDV}}}_{\text{HDV spillover}},$$

where the HDV spillover term

$$\eta_i^{\text{HDV}} = \mathbb{E} \left[\int_0^T \sum_{k \neq i} \left| \Psi_k(\zeta_t^{\phi'}(X_{k,t})) - \Psi_k(\zeta_t^\phi(X_{k,t})) \right| dt \right]$$

captures the cost variation, integrated over the horizon, that vehicle i 's deviation transmits to the other agents through the HDV field.

Bounding η_i^{HDV} requires understanding how the HDV field responds to a unilateral CAV deviation. Under a Lipschitz assumption $|\Psi_k(\zeta_t^{\phi'}(X_{k,t})) - \Psi_k(\zeta_t^\phi(X_{k,t}))| \leq L_\Psi |\zeta_t^{\phi'}(X_{k,t}) - \zeta_t^\phi(X_{k,t})|$, and since $\zeta_t = \omega * \rho_t^{\text{HDV}}$, controlling η_i^{HDV} reduces to controlling

$$\|\rho_t^{\text{HDV},\phi'} - \rho_t^{\text{HDV},\phi}\|_{L_x^1}$$

over $t \in [0, T]$. This is a stability estimate for a scalar conservation law under a perturbation of its flux, since the unilateral deviation of a CAV enters the LWR equation only through $\rho_t^{\text{CAV},\phi}$, which appears inside the flux, $V_H(\rho^{\text{HDV},\phi}(x, t) + \rho_t^{\text{CAV},\phi}(x))$.

5.2 The Colombo–Mercier–Rosini stability theorem

To control the HDV-field perturbation we appeal to the stability theorem of Colombo, Mercier, and Rosini (2008), *Stability estimates on general scalar balance laws*, Theorem 2.4. Applied to our LWR-type equation, the theorem yields, under regularity hypotheses H1–H3 on the flux,

$$\sup_{t \in [0, T]} \|\rho_t^{\text{HDV}, \phi'} - \rho_t^{\text{HDV}, \phi}\|_{L_x^1} \leq \int_0^T E_i(t) dt \leq C_i^{\text{HDV}},$$

for a constant C_i^{HDV} depending on the size of the perturbation in the flux and on the regularity of V_H and the CAV densities. Plugging this estimate into the spillover bound and using the kernel smoothing,

$$\eta_i^{\text{HDV}} \leq \|\omega\|_{L^\infty} \left(\sum_{k \neq i} L_{\Psi_k} \right) \int_0^T E_i(t) dt \leq T(N-1)L_\Psi \|\omega\|_{L^\infty} C_i^{\text{HDV}},$$

where the second inequality uses the uniform bound $L_{\Psi_k} \leq L_\Psi$.

5.3 Main bound

Combining the two pieces yields the central result of the project: under a Lipschitz assumption on Ψ and provided the LWR PDE solution map is L^1 -stable with respect to perturbations in the induced CAV density,

$$\alpha = |\Delta_i \Phi^{\text{dep}} - \Delta_i J_i| \leq T \|K\|_\infty \max_{i \in [N]} \sum_{j \neq i} |\lambda_{ij} - \lambda_{ji}| + T(N-1)L_\Psi \|\omega\|_{L^\infty} C_i^{\text{HDV}}.$$

The new HDV spillover correction is governed by the Lipschitz constant L_Ψ of the exposure penalty, the size of the smoothing kernel ω , and the Colombo–Mercier–Rosini PDE stability constant C_i^{HDV} .

5.4 Regularity assumptions

The Colombo–Mercier–Rosini bound holds under regularity hypotheses H1–H3. In our setting these translate to two requirements: the HDV velocity V_H must be smooth enough that derivatives of $\rho^{\text{HDV}, \phi}$ exist and remain bounded; and the CAV densities $\rho_t^{\text{CAV}, \phi}$ and $\rho_t^{\text{CAV}, \phi'}$ must be regular in x and t , since differentiating the flux $f^\phi(t, x, u) = u V_H(u + \rho_t^{\text{CAV}, \phi}(x))$ produces derivatives of $\rho_t^{\text{CAV}, \phi}$. Verifying that our specific LWR coupling satisfies H1–H3 marks the current iteration of our research progress. The simulation in the following section is intended to provide empirical evidence that the bound holds for the parameter ranges we use.

6 Fixed-Point Iteration Algorithm

We now look to empirically validate the upper bound on α and whether the Colombo–Mercier–Rosini assumption is appropriate for our formulation. The first step is to simulate the coupled CAV–HDV setting and verify that it behaves correctly before considering perturbations to the HDV field.

A practical limitation of moving from the pure-CAV setting to the coupled CAV–HDV setting is that Di’s decentralized policy-gradient scheme no longer applies directly. The original algorithm decomposes Φ into per-agent contributions and updates each ϕ_i using locally computable gradients

of J_i . This decomposition relies on each agent’s cost being self-contained. This is no longer true in our coupled system. The HDV field $\rho^{\text{HDV},\phi}$ depends on the full profile ϕ , and it appears in every other agent’s cost, so a single agent’s gradient step requires evaluating the HDV field that the entire population has collectively shaped.

To resolve this, we introduce a fixed-point algorithm that alternates between optimizing CAV policies under a fixed HDV field, and then re-solving the LWR PDE under these updated policies. After many iterations, we look to attain a pair $(\phi^*, \rho^{\text{HDV},*})$ such that ϕ^* is the optimal CAV profile under field $\rho^{\text{HDV},*}$, and $\rho^{\text{HDV},*}$ is the HDV density induced by ϕ^* via the LWR equation. The algorithm proceeds as follows. At outer iteration k :

1. Given the current profile $\phi^{(k)}$, simulate CAV trajectories to compute $\rho_t^{\text{CAV},\phi^{(k)}}$, then solve the LWR PDE forward in time to obtain $\rho^{\text{HDV},(k)}$.
2. Holding $\rho^{\text{HDV},(k)}$ fixed, run 100 steps of gradient descent on each agent’s cost J_i to obtain an inner-update profile ϕ^{inner} .
3. Update with damping to prevent solutions from oscillating and instead converge to a fixed point: $\phi^{(k+1)} = (1 - \beta)\phi^{(k)} + \beta\phi^{\text{inner}}$, with $\beta = 0.5$.
4. Resimulate trajectories under $\phi^{(k+1)}$ to obtain $\rho^{\text{HDV},(k+1)}$.
5. Stop when $\|\phi^{(k+1)} - \phi^{(k)}\|_{\text{MSE}} < 10^{-6}$ and $\|\rho^{\text{HDV},(k+1)} - \rho^{\text{HDV},(k)}\|_{L^1} < 5 \times 10^{-4}$ for five consecutive iterations, or after 1000 outer iterations.

7 Numerical Experiments

We implemented the fixed-point algorithm on a periodic ring road of length $L = 1$ over horizon $T = 1$ with $N = 3$ CAVs whose target positions are $z_i = (0.25, 0.5, 0.75)$ and initial positions $X_{i,0} = (0.1, 0.5, 0.9)$. Each CAV’s policy $\phi_i(t, x)$ is parameterized as a small multilayer perceptron with two hidden layers of 32 units and tanh activations, mapping $(t, x) \in [0, T] \times \mathbb{T}$ to a scalar velocity. Raw outputs are clipped via a soft action bound $a_{\max} \tanh(a/a_{\max})$. Policies are trained by manual backpropagation through time with analytic gradients.

The HDV field is discretized on 200 cells and integrated with a Lax–Friedrichs scheme using a Greenshields velocity $V_H(\rho) = v_{\max}(1 - \rho/\rho_{\text{jam}})$ with $v_{\max} = \rho_{\text{jam}} = 1$. The CAV kernel-density estimate uses a periodic Gaussian K_h of bandwidth $h_{\text{CAV}} = 0.04$, and the HDV exposure kernel ω uses bandwidth 0.03. Cost-function parameters are $f_i = 0.1 a_i^2$, $g_i = 10 (X_{i,T} - z_i)^2$ with periodic distance, $\lambda_{ij} = 0.5$, and $\Psi_i(\zeta) = \zeta^2$ scaled by $\psi_{\text{coef}} = 1$.

We test three HDV initial conditions: a uniform density $\rho_0 \equiv 0.3$, a single Gaussian bump of peak 0.6 centered at $x = 0.5$, and two Gaussian bumps of peak 0.5 at $x = 0.25$ and $x = 0.75$. The two-bumps and bump configurations are intended to act as obstacles that the CAVs must navigate.

7.1 Results

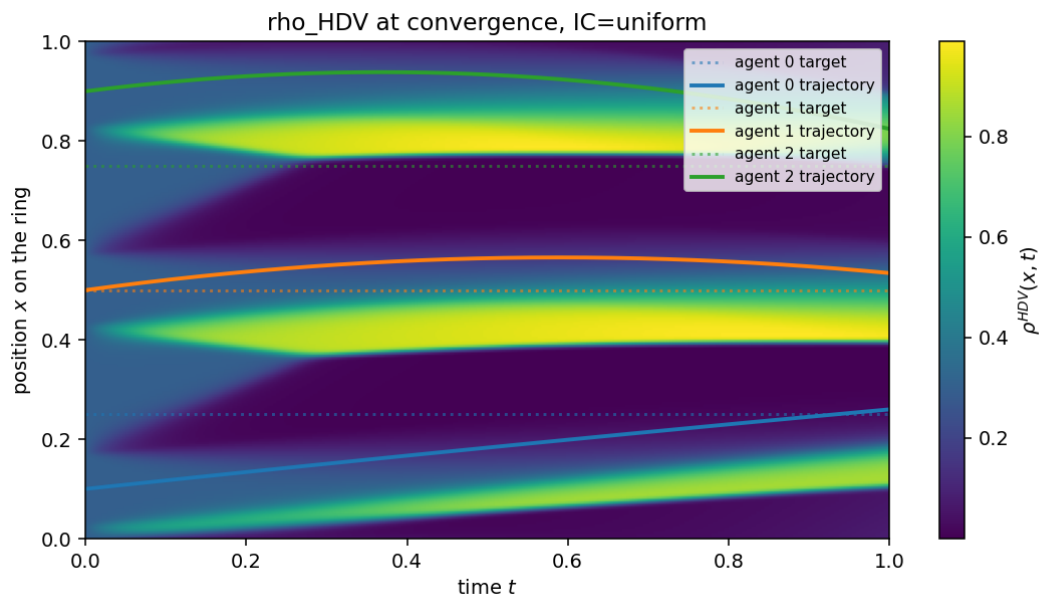


Figure 1: Uniform IC

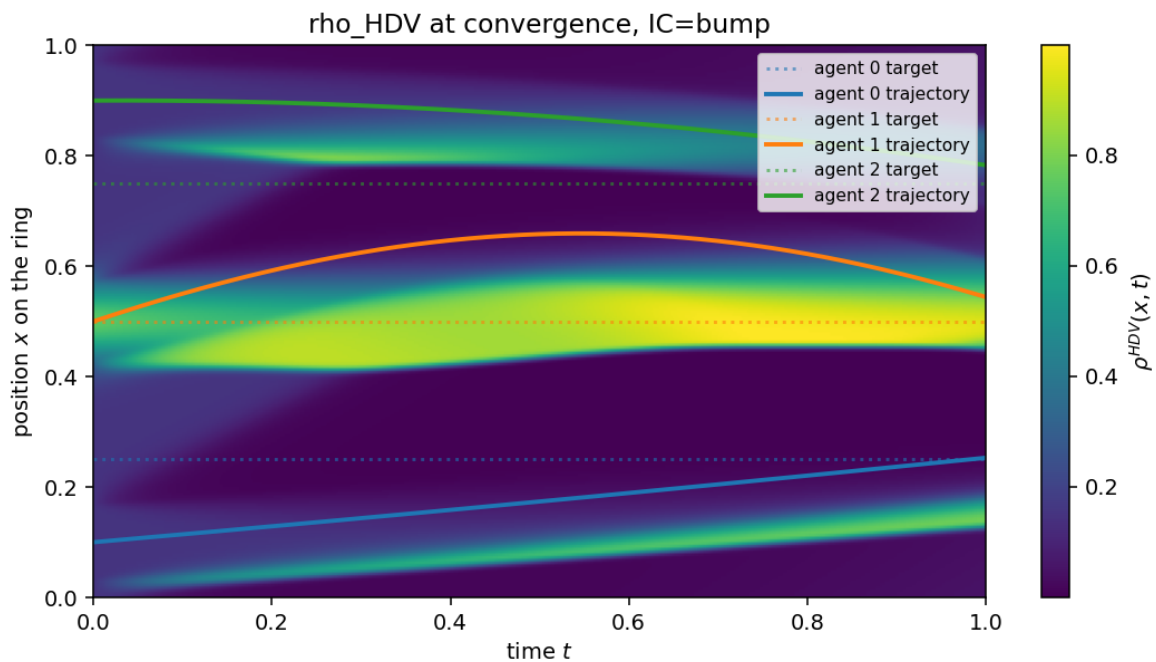


Figure 2: Bump IC

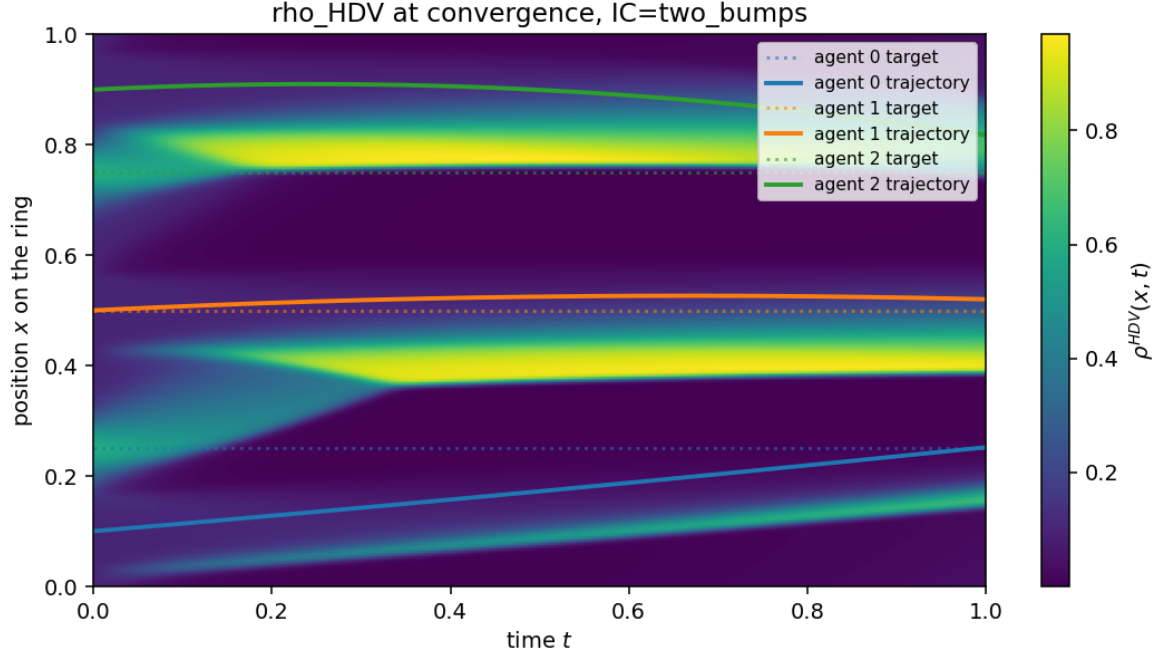


Figure 3: Two-bumps IC

I presented the figures above in the most recent meeting and there were several concerns about the simulation output. First, the HDV field is exhibiting behavior that cannot be explained by the model as written. Even with CAVs effectively negligible (only three sparse vehicles), the LWR equation alone should not produce the patterns observed. For instance, in the figure with uniform initial density, no congestion should form, yet the heatmaps show congestion arising spontaneously, and HDV density appears to increase in regions not aligning with conservation. We expect there to be an issue in the way the LWR solver is configured. The next steps to address this problem would be to first determine whether CFL condition has been verified and running the LWR equation in isolation (no CAV coupling) to confirm whether the solver behaves correctly before incorporating the coupled dynamics.

The second concern arises from the CAV–HDV coupling itself. The Gaussian smoothing kernel used to compute ρ^{CAV} has bandwidth $h_{\text{CAV}} = 0.04$, which produces too wide influence on the HDV field. A single CAV appears to affect traffic over a region of width 0.2, which is too large given the sparsity of the CAVs. There are two approaches to this problem. One is to drop the Gaussian smoothing in favor of the empirical density itself, which we would derive from a sum of Dirac masses at the CAV positions. Another method is to replace the Gaussian with a rectangular moving-bottleneck kernel whose support is the vehicle length and its height is tied to a fraction of the critical density. We will pursue the empirical density approach first, since it best aligns with what we have been doing up to this point. A final concern is that on the heatmaps some CAV trajectories run backwards on the ring, which is indicative of insufficient action constraints during policy training. It is recommended to clip the velocity to $[0, v_{\text{max}}]$ during the gradient-descent updates.

Resolving these numerical and modeling issues is the immediate next step before we can make any further progress on validating the α bound. Looking forward, the planned summer extension of this

will continue to address the Colombo–Mercier–Rosini regularity conditions in our CAV–HDV setting by simulating the impact on the HDV field when a vehicle unilaterally deviates. Once we obtain significant results for our toy example, we would like to scale up the simulations to a moderate/large number of CAVs and examine the interactions in that setting. By the end of the summer, we aim to publish a paper built on the semester-long progress demonstrated through this document.

Bibliography of papers examined over this semester

- Di, X., Hu, A., Wang, Z., Zhang, Y. *α -Potential Games for Decentralized Control of Connected and Automated Vehicles*. arXiv:2512.05712, 2025.
- Guo, X., Li, X., Maheshwari, C., Sastry, S., Wu, M. *Markov α -potential games*. IEEE Trans. Autom. Control, 2025.
- Huang, K., Di, X., Du, Q., Chen, X. *Scalable traffic stability analysis in mixed-autonomy using continuum models*. Transp. Res. Part C, 2020.
- Colombo, R. M., Mercier, M., Rosini, M. D. *Stability estimates on general scalar balance laws*. 2008.